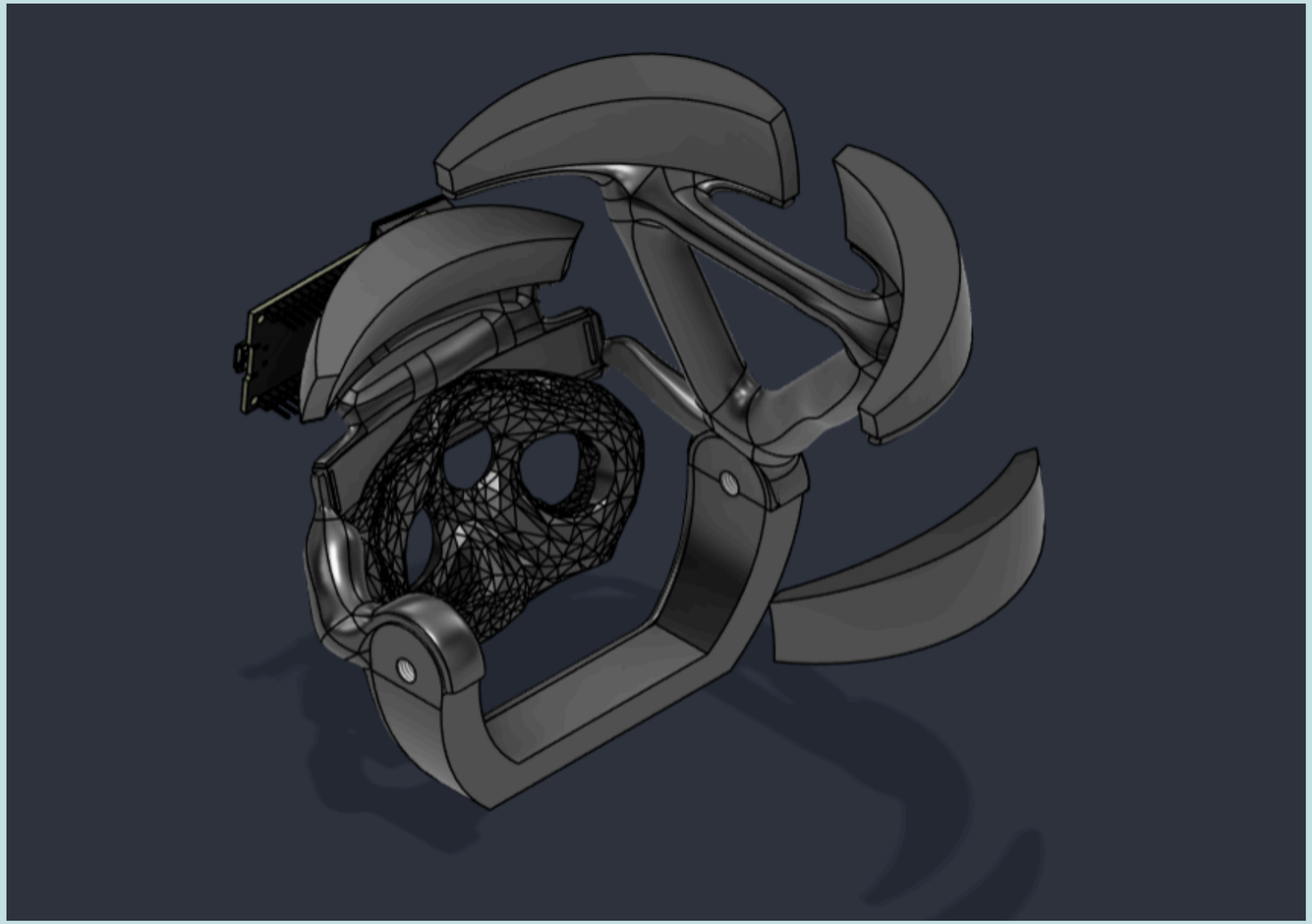
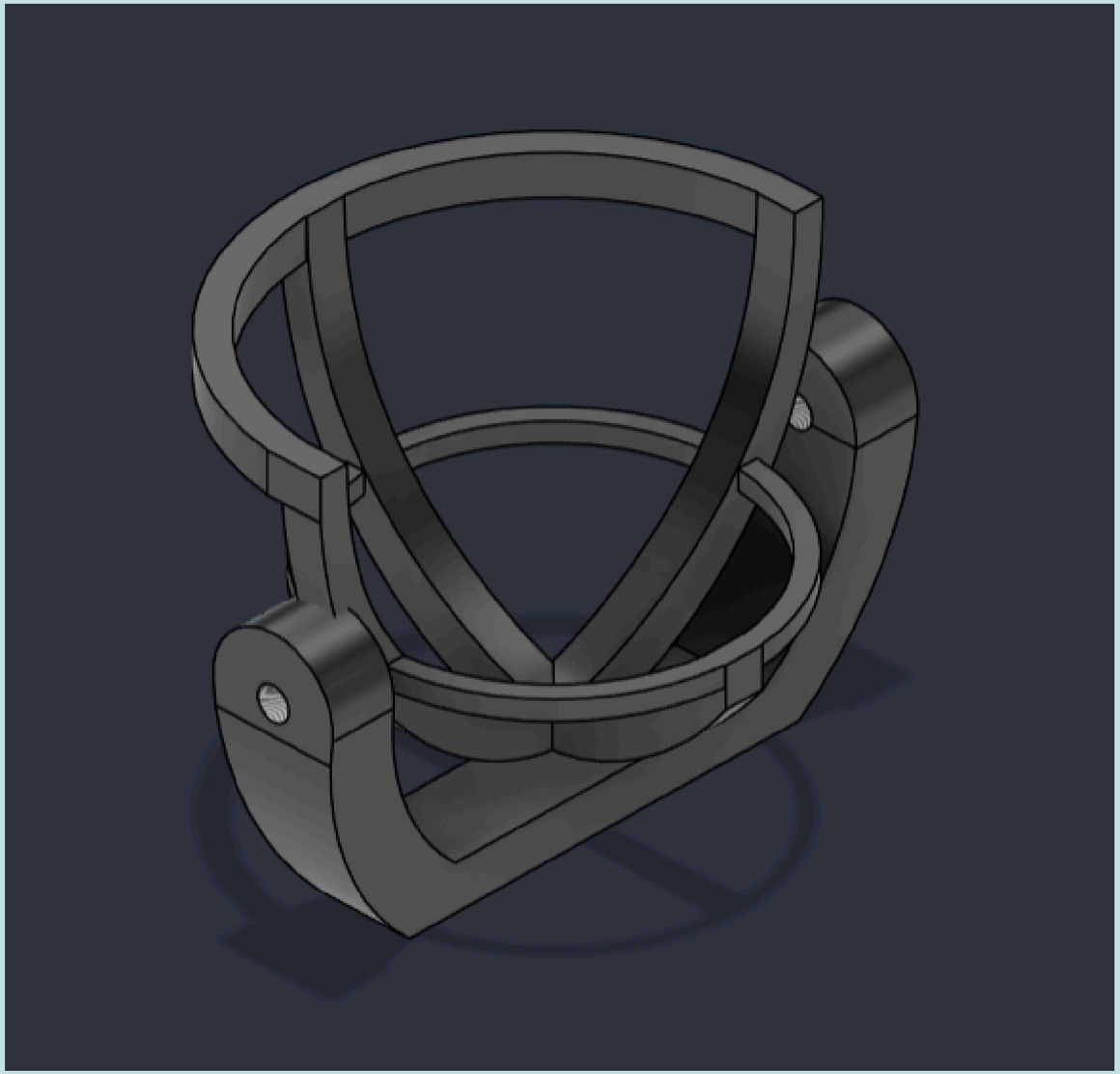
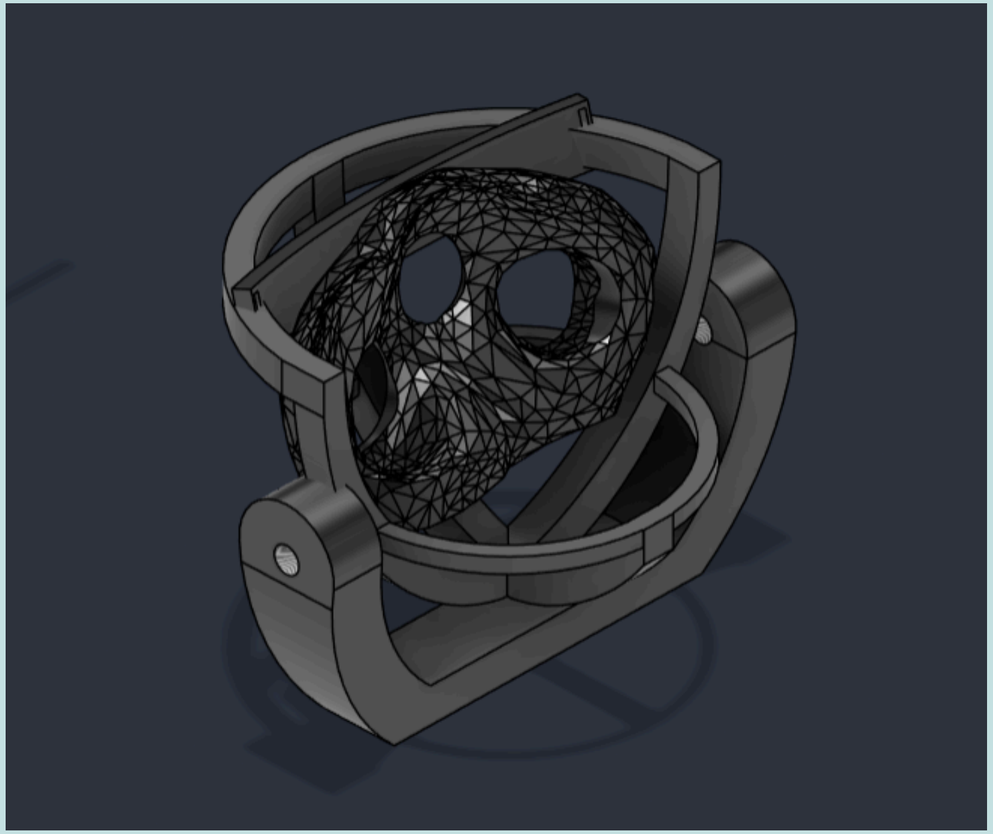
Grupo 14

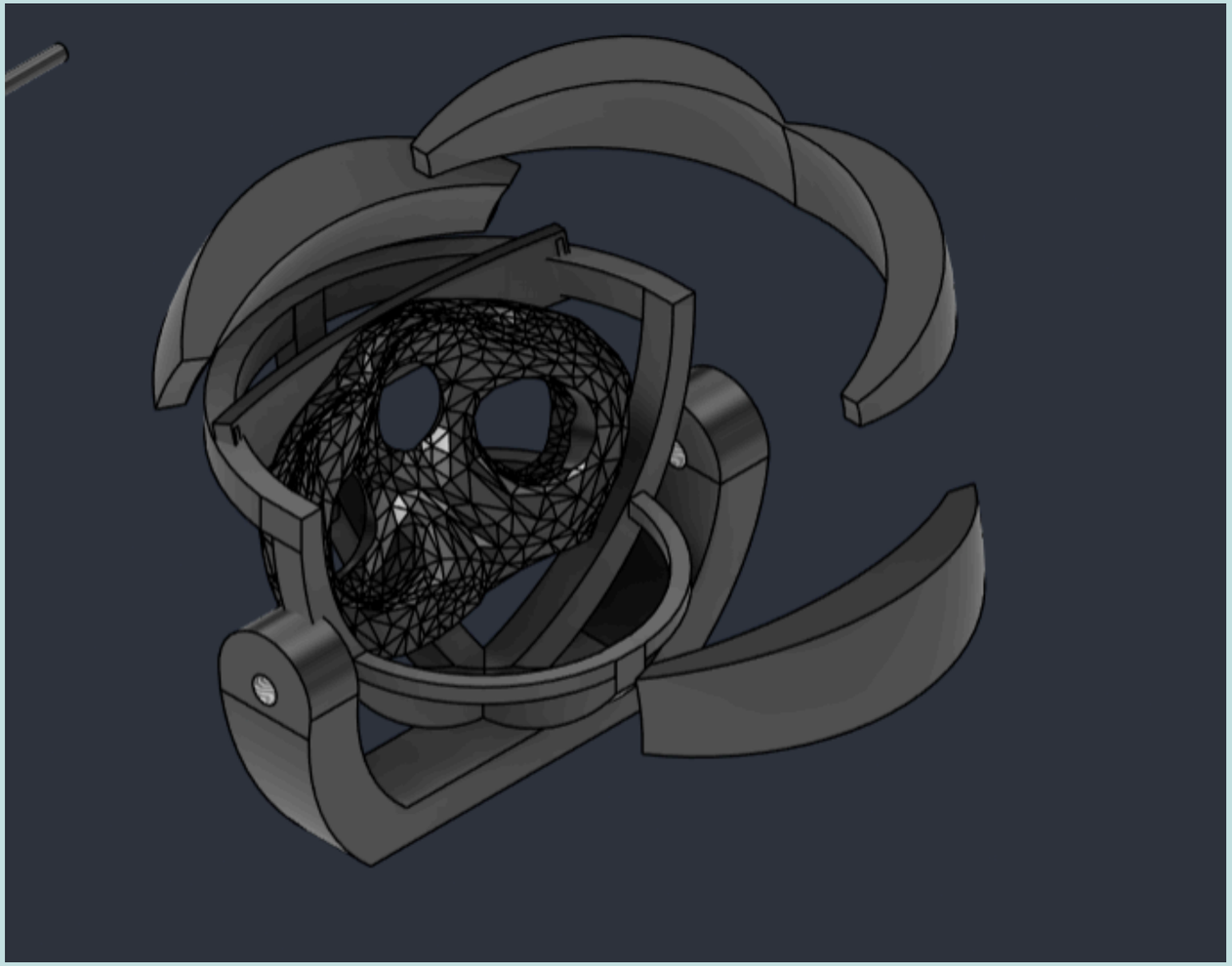
ENTREGABLE 10

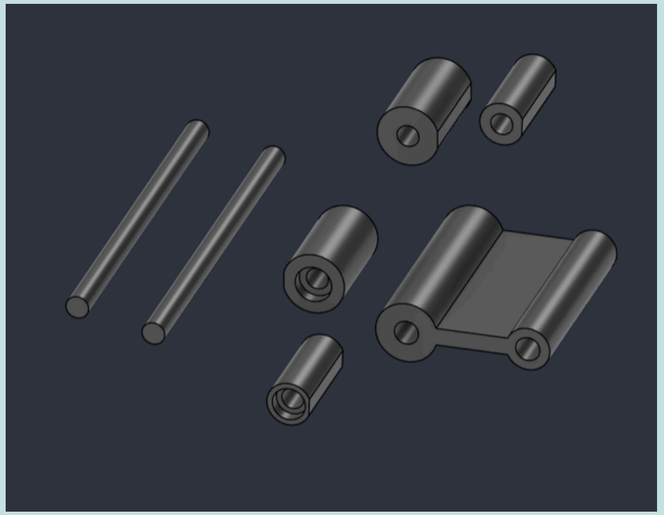


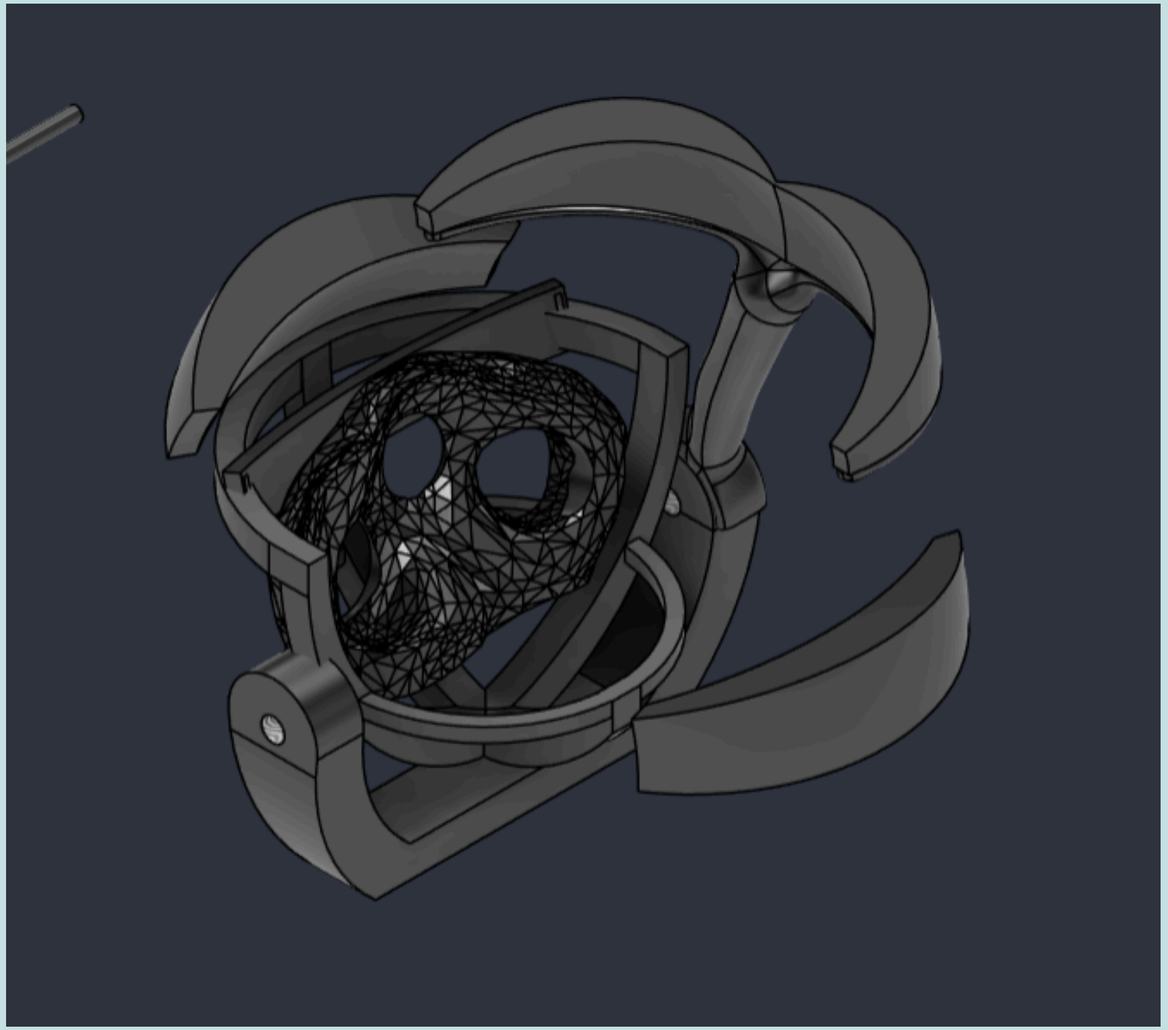




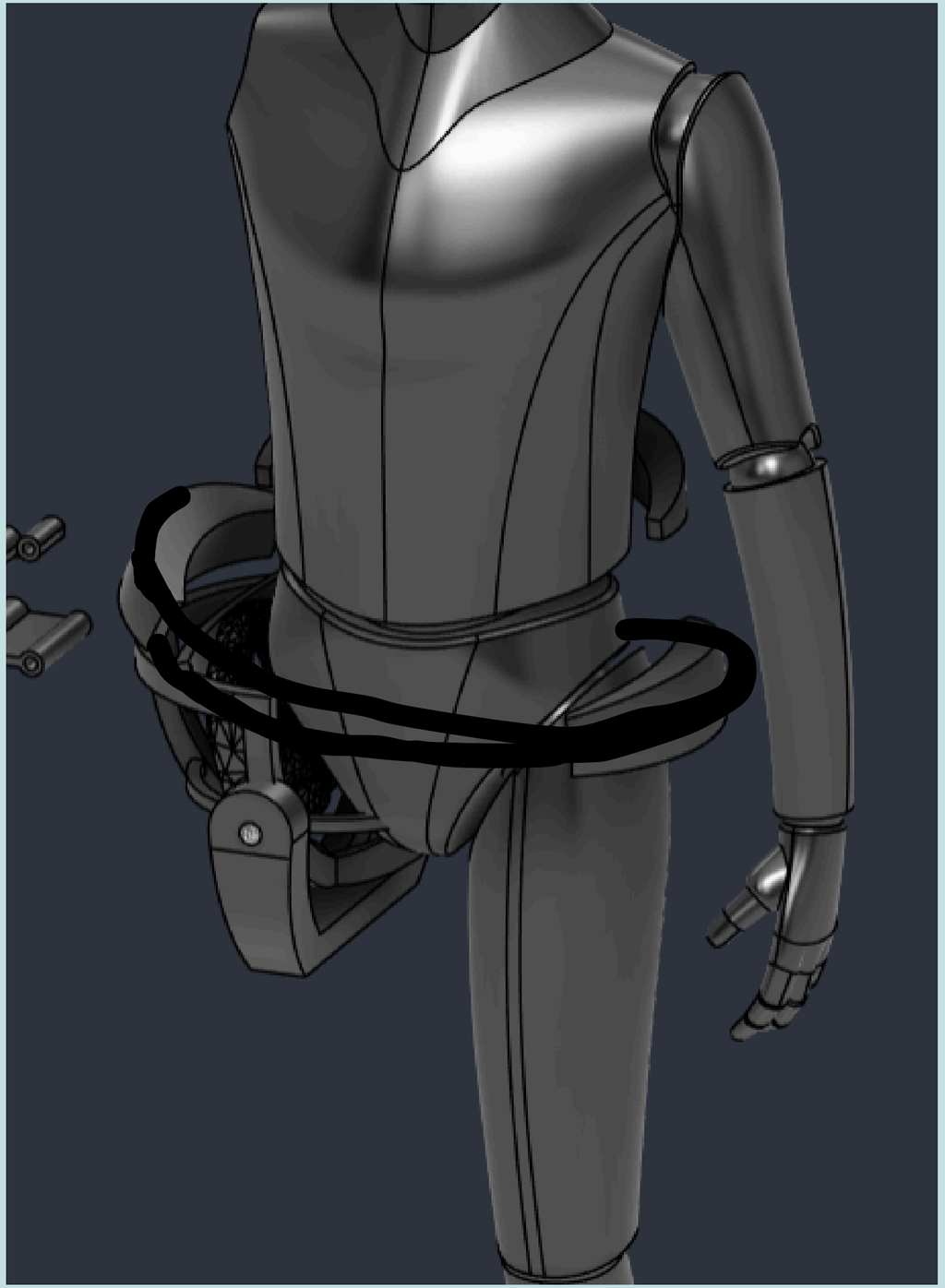


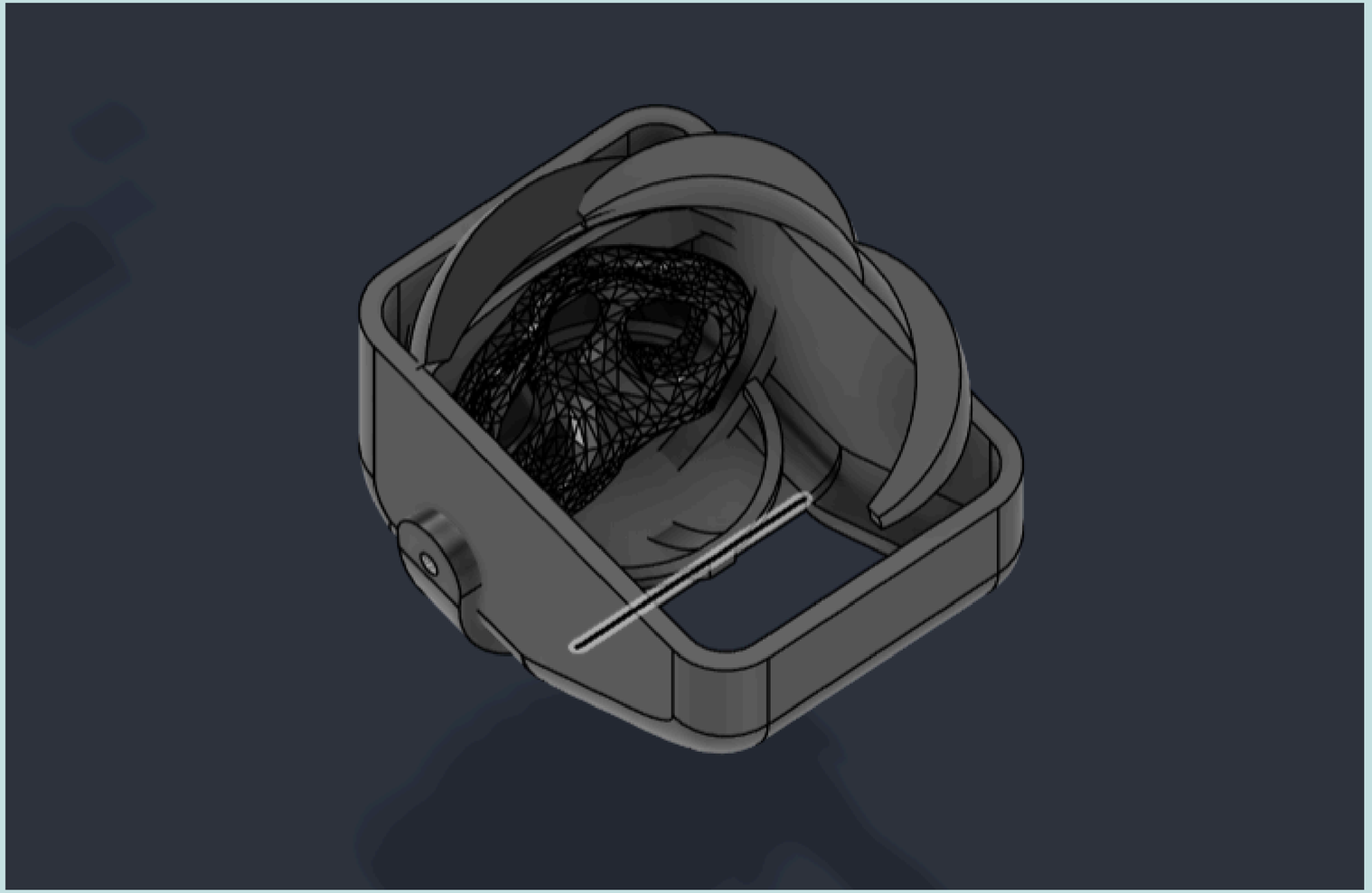


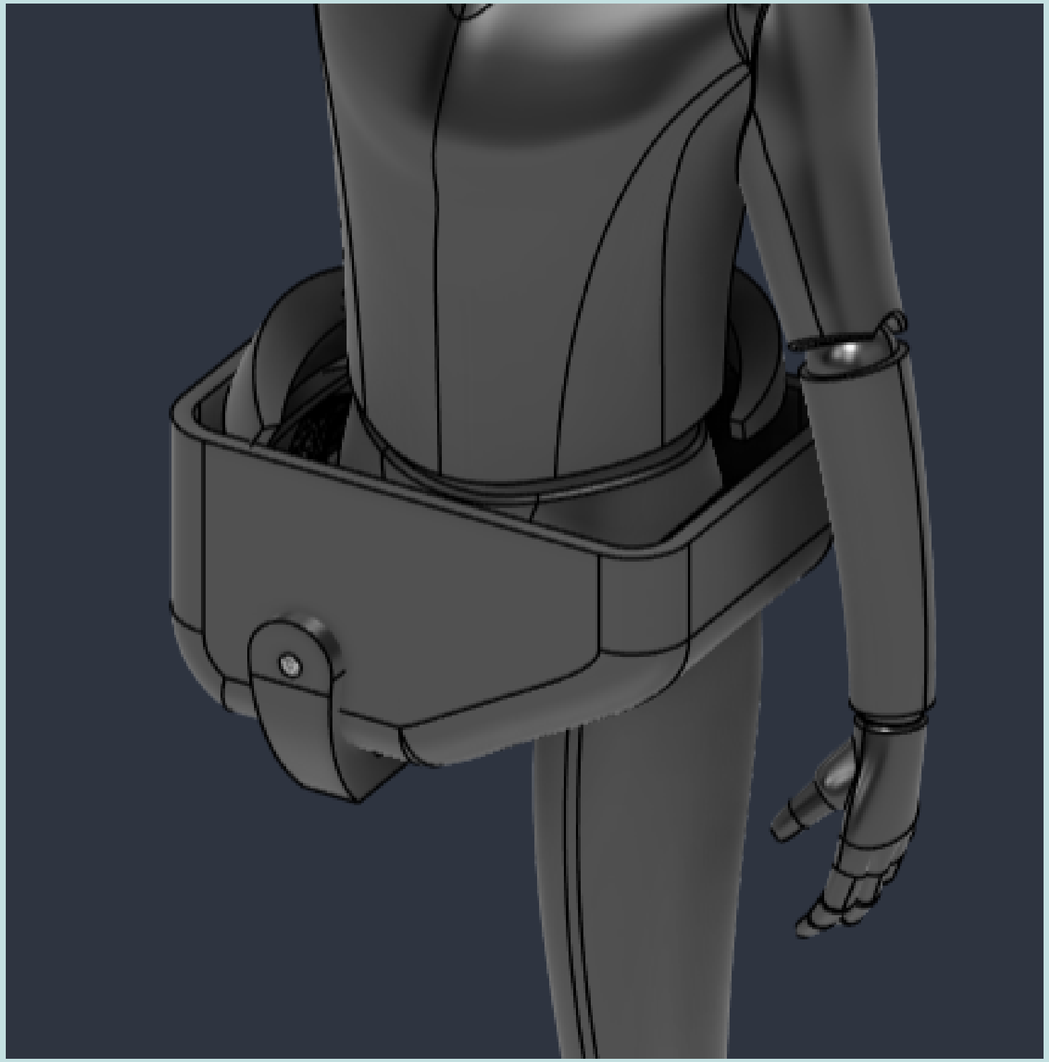




Doc One Shape:
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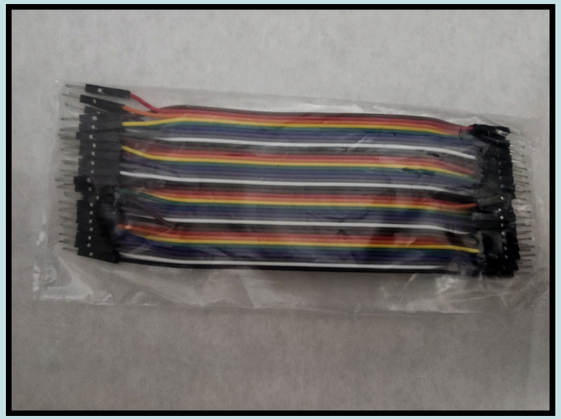




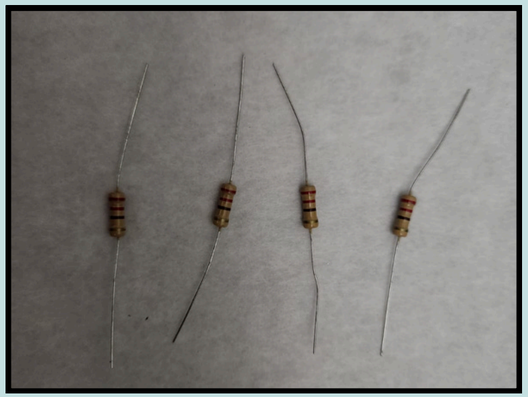
ELECTRONICA Y SENSORES

COMPONENTES

Cables jumper

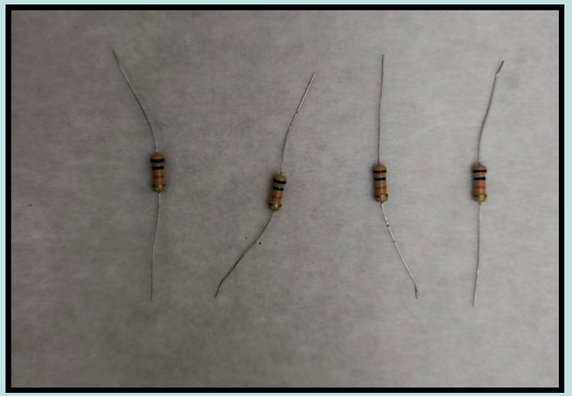


Resistencias de 10k y 120 ohm

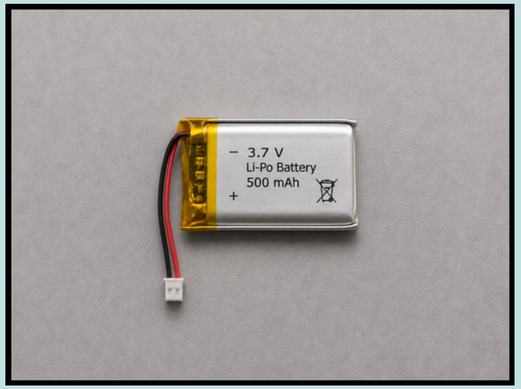


Sensores FSR





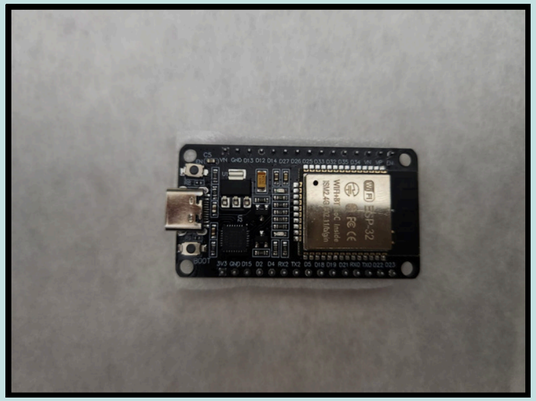
Bateria litio



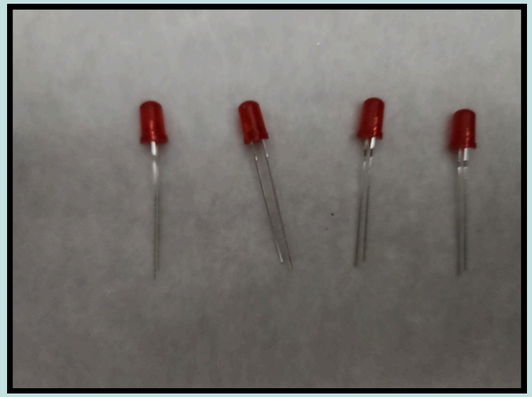
Boton encender/apagar



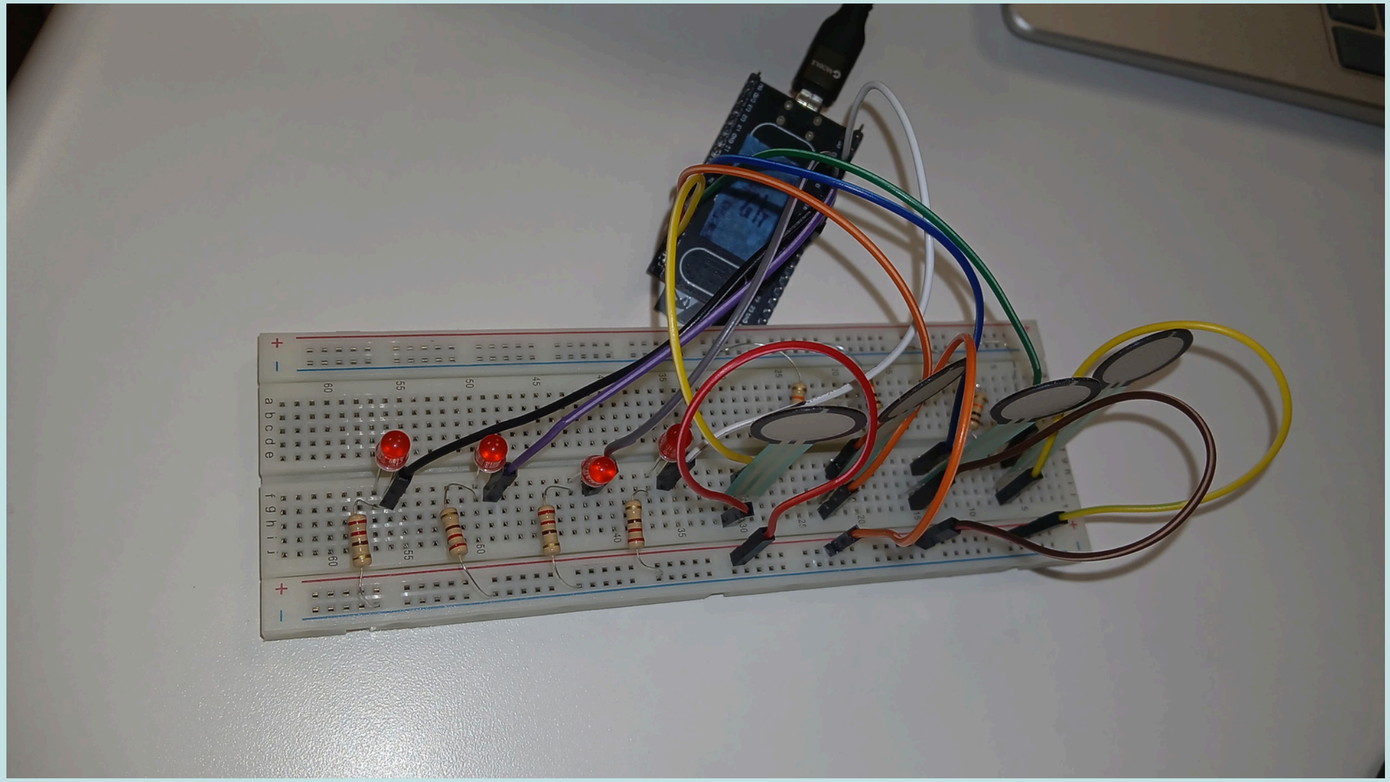
ESP32



LED



Ensamble del sistema electrónico



CÓDIGO

```
#include "BluetoothSerial.h"

BluetoothSerial SerialBT;

// Pines de los 4 sensores FSR (ADC1)
const int fsr1Pin = 36;
const int fsr2Pin = 39;
const int fsr3Pin = 34;
const int fsr4Pin = 35;

// Pines de los 4 LEDs
const int led1Pin = 2;
const int led2Pin = 4;
const int led3Pin = 5;
const int led4Pin = 18;

// Umbrales individuales (valores de prueba)
const int umbral1 = 600;
const int umbral2 = 500;
const int umbral3 = 650;
const int umbral4 = 450;

void setup() {
Serial.begin(115200);
// Configuración de LEDs
pinMode(led1Pin, OUTPUT);
pinMode(led2Pin, OUTPUT);
pinMode(led3Pin, OUTPUT);
pinMode(led4Pin, OUTPUT);

// Inicialización de Bluetooth
SerialBT.begin("EncajePelvico_Pai");

Serial.println("Bluetooth listo.");
Serial.println("Monitoreando 4
sensores FSR.");
}
void loop() {

// 1. Lectura de sensores
int valor1 = analogRead(fsr1Pin);
int valor2 = analogRead(fsr2Pin);
int valor3 = analogRead(fsr3Pin);
int valor4 = analogRead(fsr4Pin);

// 2. Alertas independientes
digitalWrite(led1Pin, valor1 > umbral1 ? HIGH :
LOW);
digitalWrite(led2Pin, valor2 > umbral2 ? HIGH :
LOW);
digitalWrite(led3Pin, valor3 > umbral3 ? HIGH :
LOW);
digitalWrite(led4Pin, valor4 > umbral4 ? HIGH :
LOW);
// 3. Datos para App Inventor
String paqueteDatos =
String(valor1) + "," +
String(valor2) + "," +
String(valor3) + "," +
String(valor4);
// 4. Envío por Bluetooth
SerialBT.println(paqueteDatos);

// 5. Monitor serial
Serial.println("Enviando: " + paqueteDatos);

// 6. Intervalo de monitoreo
delay(200);
}
```

FUNCIONAMIENTO DEL PROGRAMA PRINCIPAL

Configuración global del sistema

prueba_rgb | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Select Board

prueba_rgb.ino

```
#include "BluetoothSerial.h"

BluetoothSerial SerialBT;

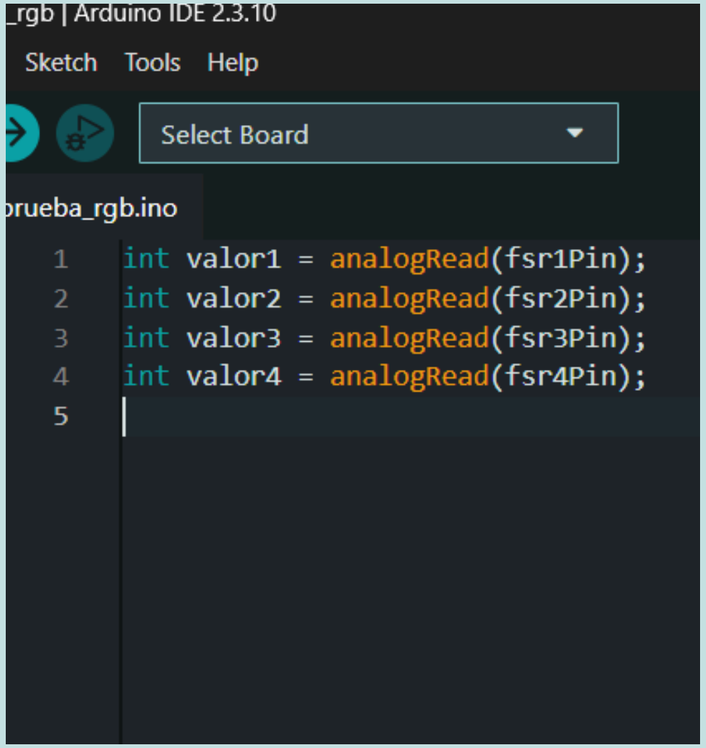
// Pines de los 4 sensores FSR
const int fsr1Pin = 36;
const int fsr2Pin = 39;
const int fsr3Pin = 34;
const int fsr4Pin = 35;

// Pines de los 4 LEDs
const int led1Pin = 2;
const int led2Pin = 4;
const int led3Pin = 5;
const int led4Pin = 18;

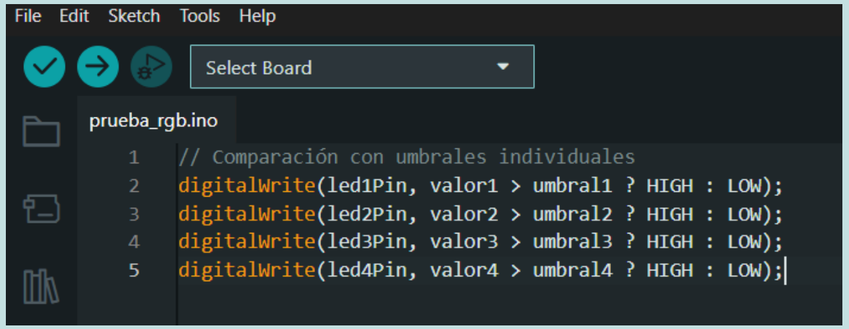
// Umbrales independientes (valores de prueba)
const int umbral1 = 600;
const int umbral2 = 500;
const int umbral3 = 650;
const int umbral4 = 450;

// Intervalo de monitoreo
const int intervalo = 200;
```

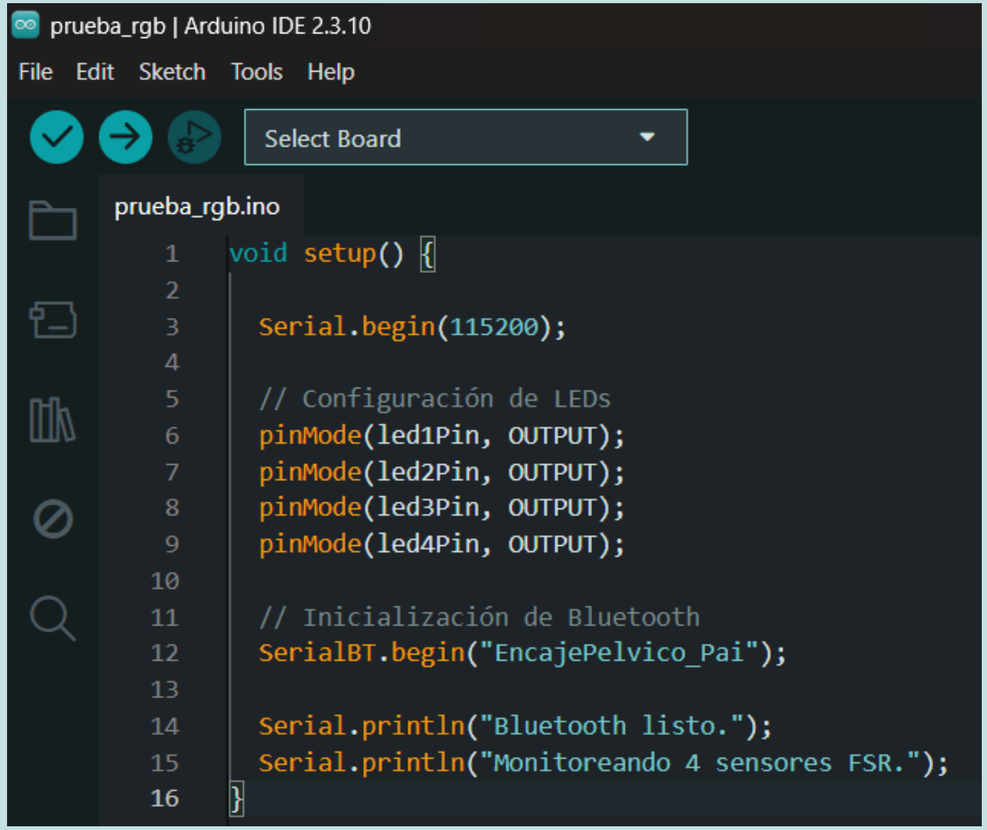
Funciones de lectura



Funciones de calibración y seguridad



Inicialización del sistema



Funcionamiento del programa principal

sketch_nov06a §

```
void loop() {

  // 1. Lectura de sensores
  int valor1 = analogRead(fsr1Pin);
  int valor2 = analogRead(fsr2Pin);
  int valor3 = analogRead(fsr3Pin);
  int valor4 = analogRead(fsr4Pin);

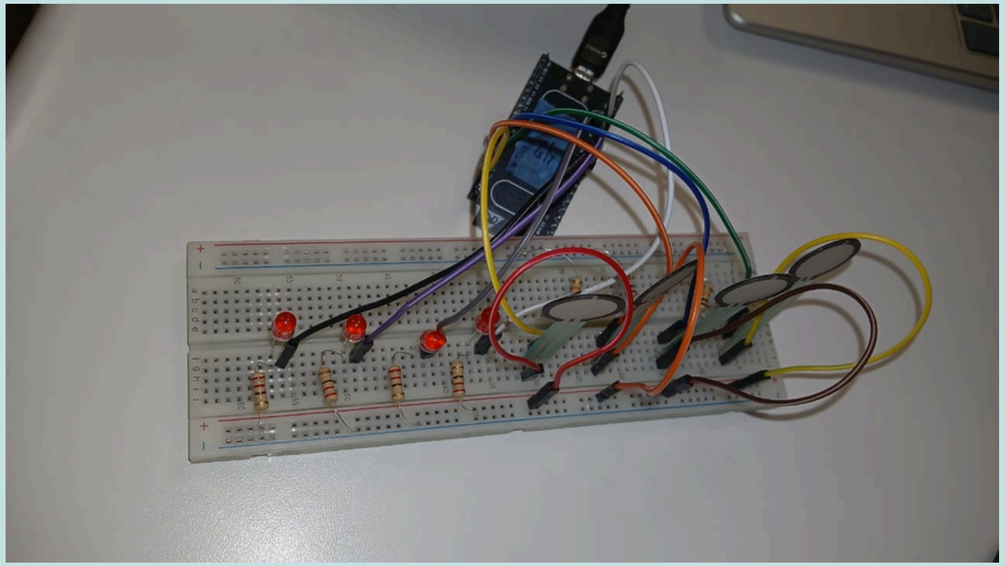
  // 2. Activación independiente de LEDs
  digitalWrite(led1Pin, valor1 > umbral1 ? HIGH : LOW);
  digitalWrite(led2Pin, valor2 > umbral2 ? HIGH : LOW);
  digitalWrite(led3Pin, valor3 > umbral3 ? HIGH : LOW);
  digitalWrite(led4Pin, valor4 > umbral4 ? HIGH : LOW);

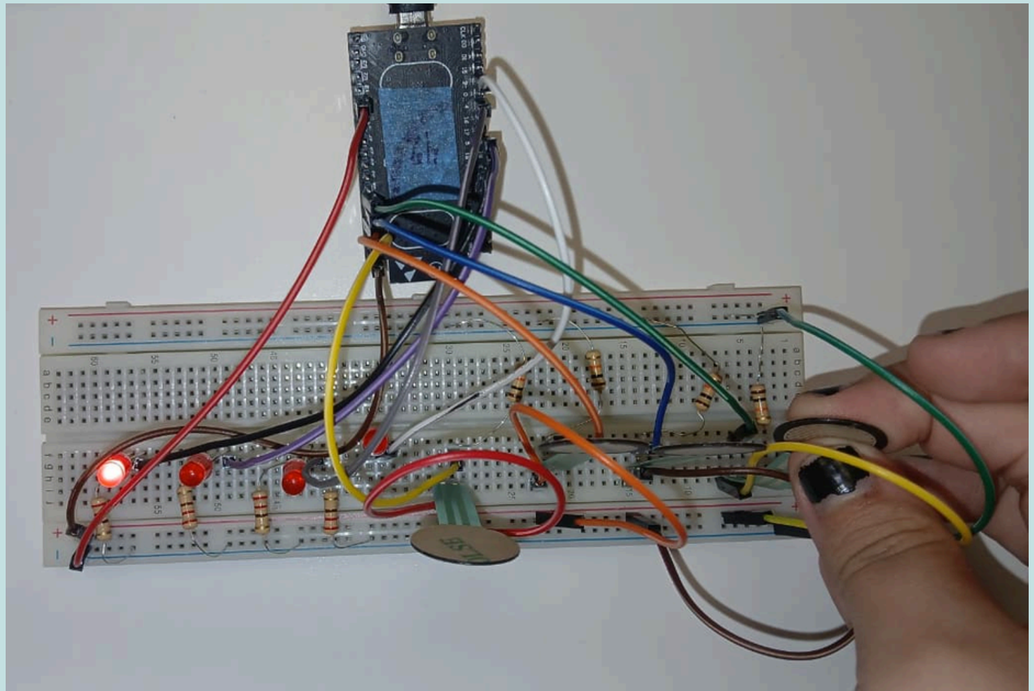
  // 3. Organización de datos
  String paqueteDatos =
      String(valor1) + "," +
      String(valor2) + "," +
      String(valor3) + "," +
      String(valor4);

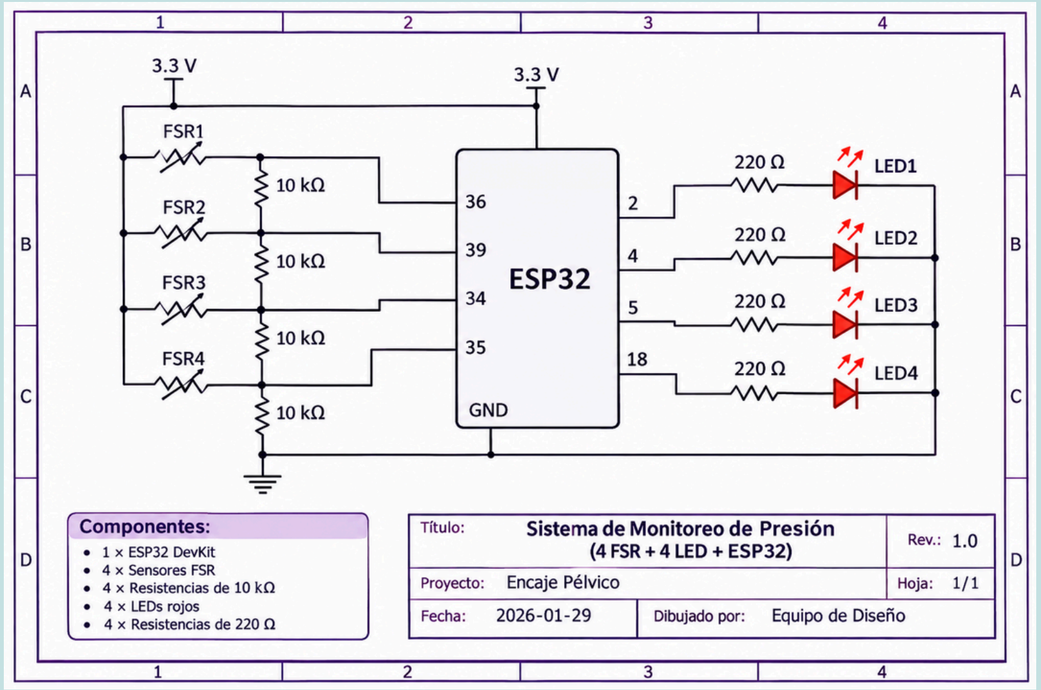
  // 4. Transmisión Bluetooth
  SerialBT.println(paqueteDatos);

  // 5. Visualización en monitor serial
  Serial.println("Enviando: " + paqueteDatos);

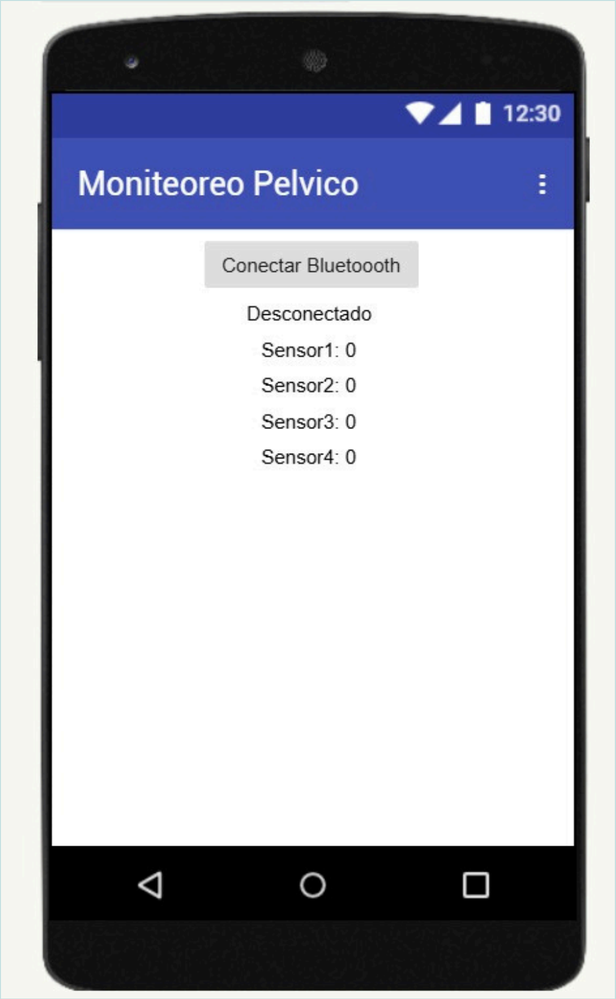
  // 6. Intervalo de monitoreo
  delay(intervalo);
}
```



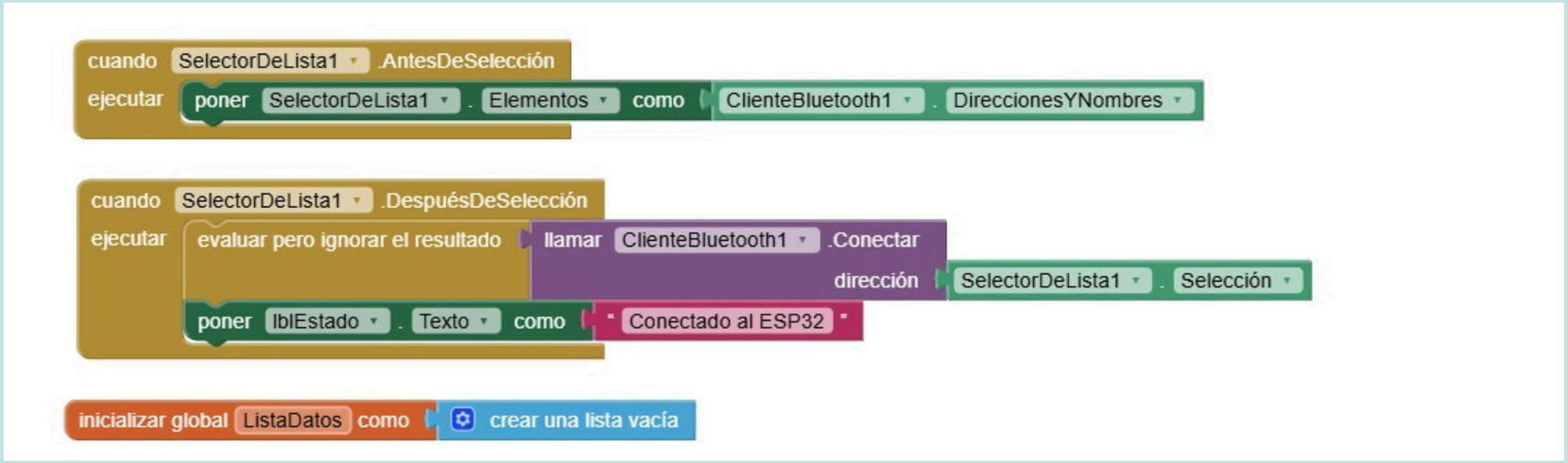


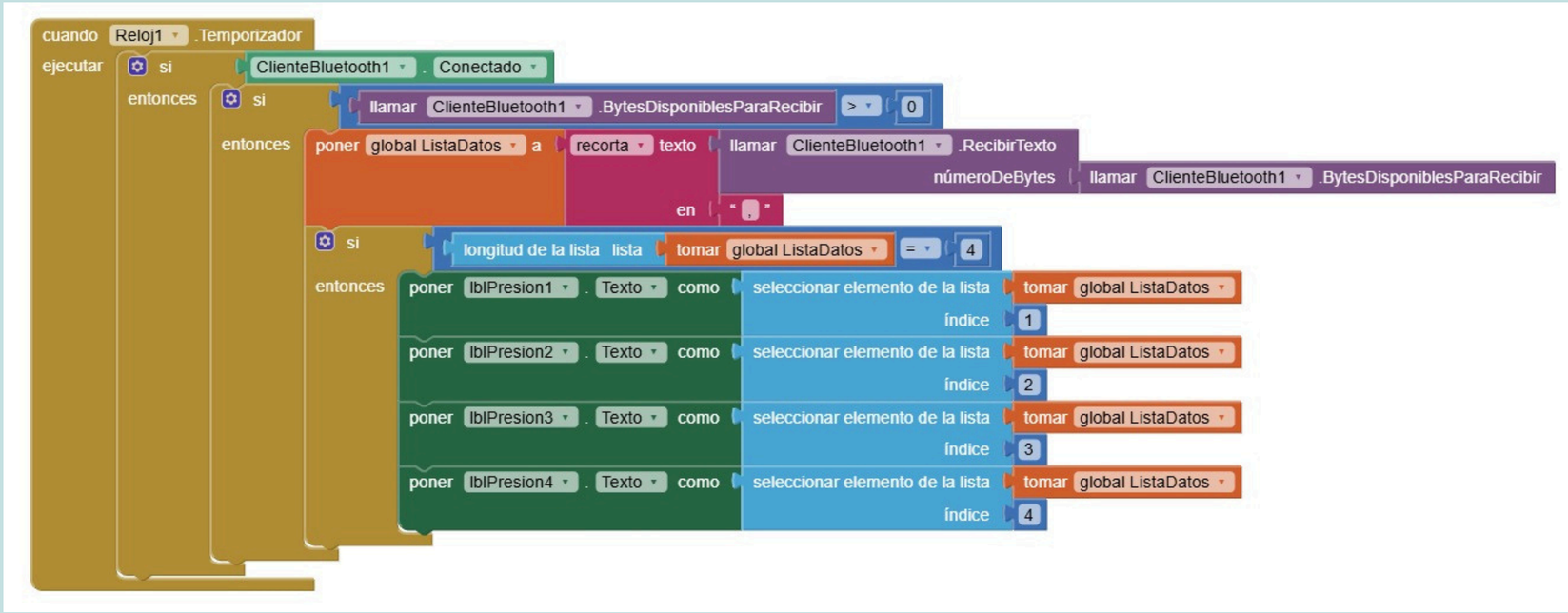

1
2
3
4
A
B
C
D
3.3 V
3.3 V
FSR1
FSR2
FSR3
FSR4
10 kΩ
10 kΩ
10 kΩ
10 kΩ
36
39
34
35
GND
ESP32
2
4
5
18
220 Ω
220 Ω
220 Ω
220 Ω
LED1
LED2
LED3
LED4
Componentes:
1 × ESP32 DevKit
4 × Sensores FSR
4 × Resistencias de 10 kΩ
4 × LEDs rojos
4 × Resistencias de 220 Ω
Título: Sistema de Monitoreo de Presión (4 FSR + 4 LED + ESP32)
Rev.: 1.0
Proyecto: Encaje Pélvico
Hoja: 1/1
Fecha: 2026-01-29
Dibujado por: Equipo de Diseño

App de monitero pelvico



Bloques de programacion en MIT App Inventor





DIAGRAMAS DE FLUJO

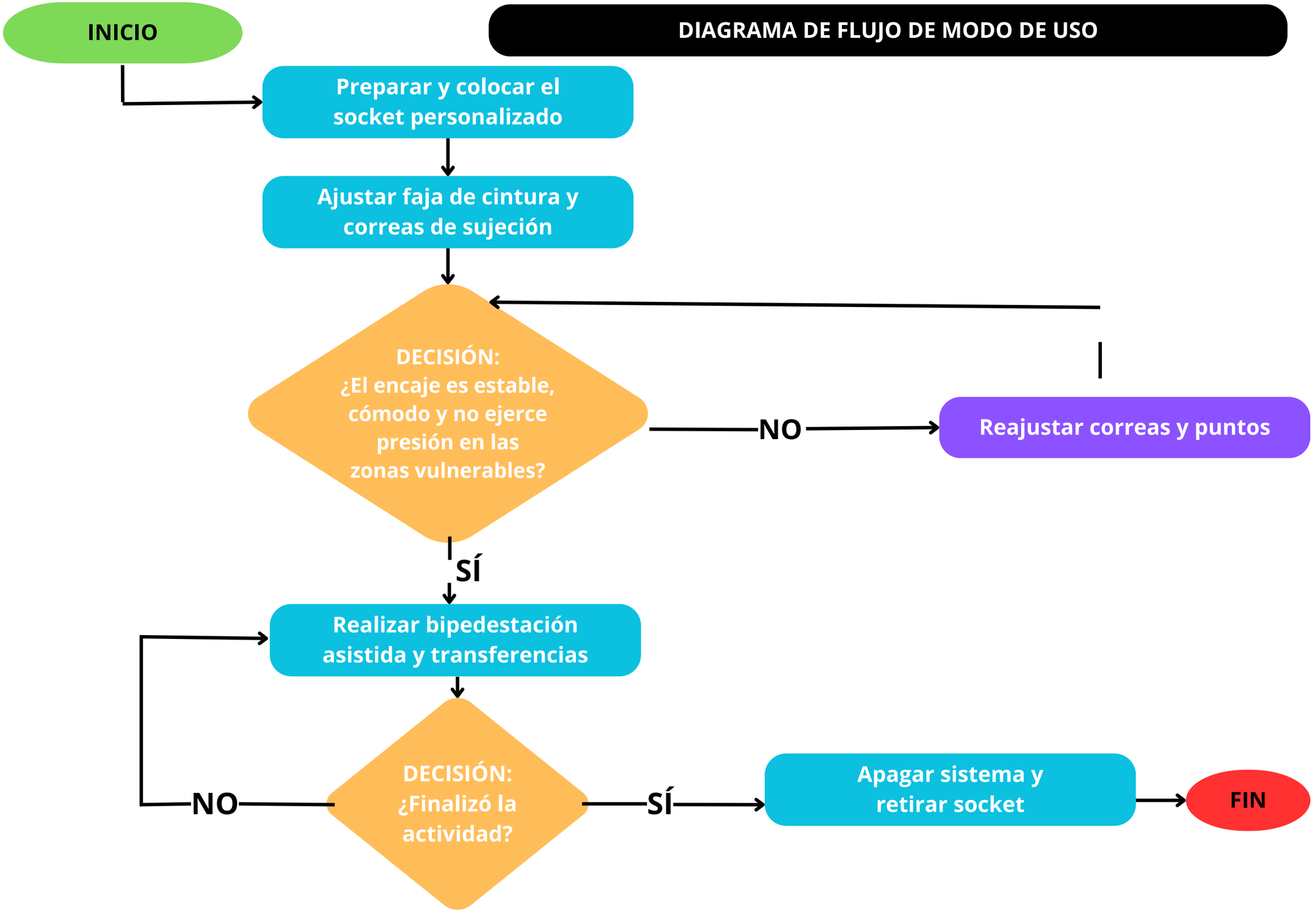

INICIO
DIAGRAMA DE FLUJO DE MODO DE USO
Preparar y colocar el socket personalizado
Ajustar faja de cintura y correas de sujeción
DECISIÓN: ¿El encaje es estable, cómodo y no ejerce presión en las zonas vulnerables?
NO
Reajustar correas y puntos
SÍ
Realizar bipedestación asistida y transferencias
DECISIÓN: ¿Finalizó la actividad?
NO
SÍ
Apagar sistema y retirar socket
FIN

DIAGRAMA DE FLUJO DE MODO DE FUNCIONAMIENTO DE LOS SENSORES

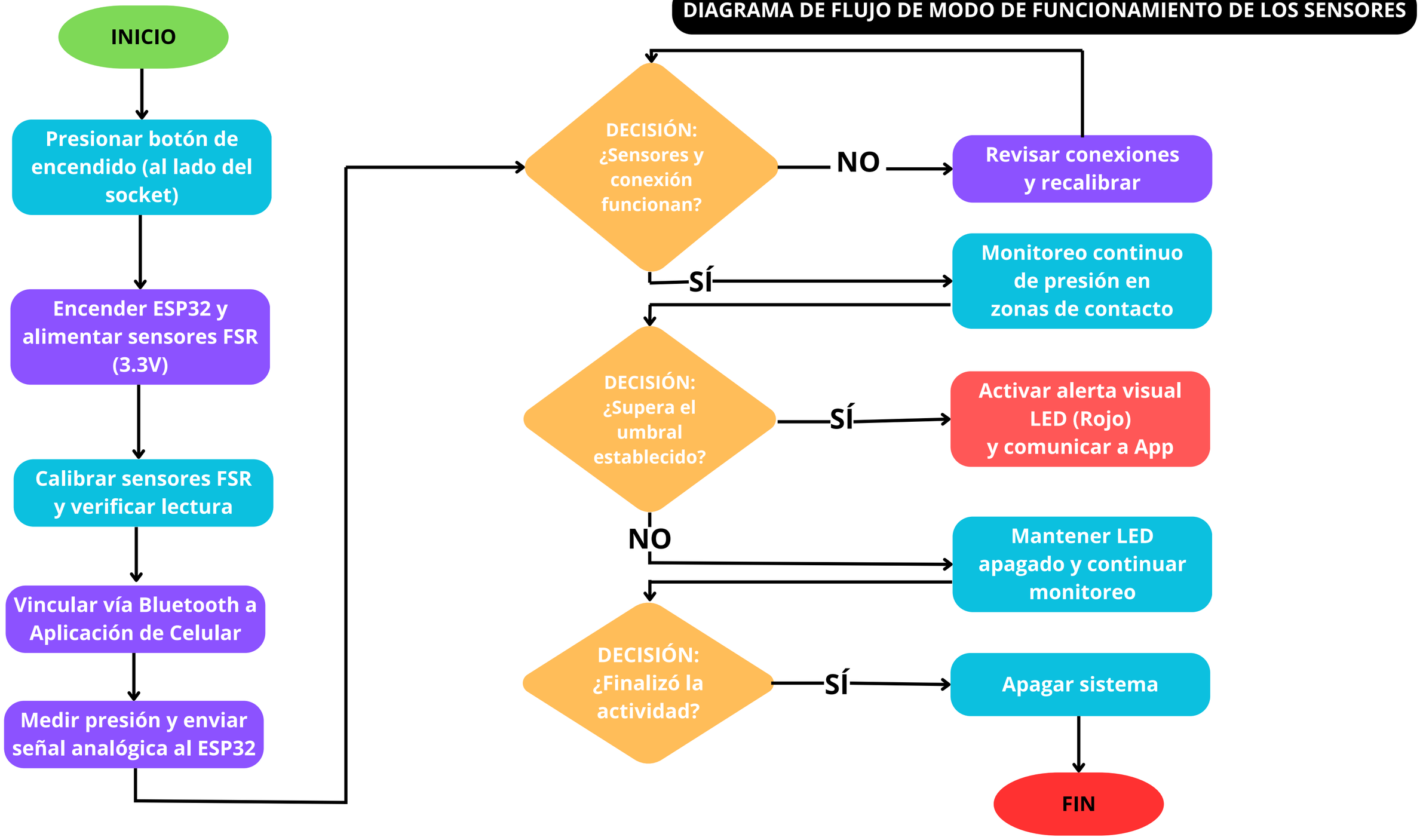
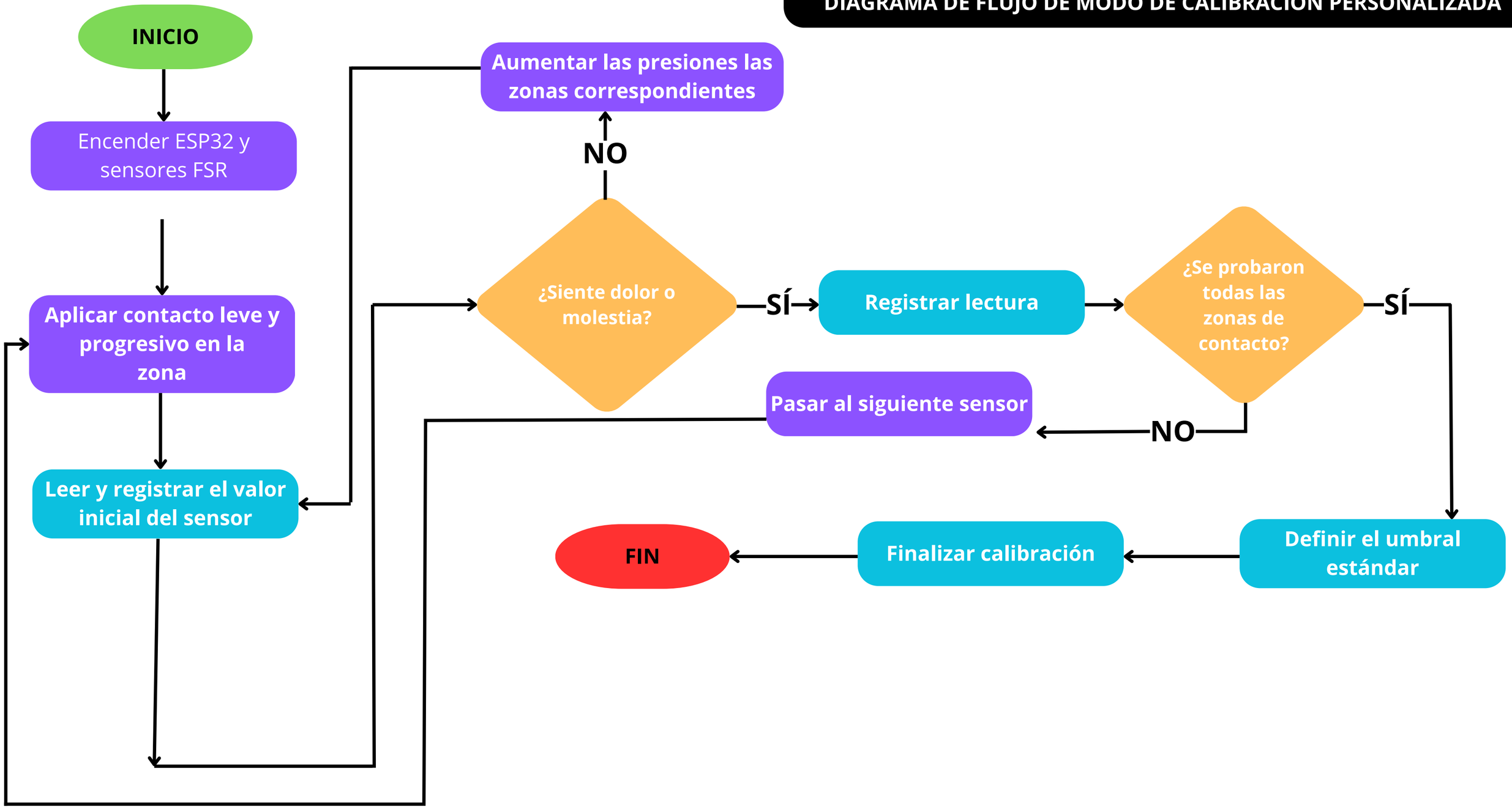


DIAGRAMA DE FLUJO DE MODO DE CALIBRACION PERSONALIZADA



Equipo 14

¡Muchas gracias por la atención!

